

NaCl-zwitterion enhanced chitosan-based hydrogels with super compressibility, conductivity, and stretchability as wearable strain and pressure sensors

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ABSTRACT

Conductive hydrogels, combining flexibility with electrical conductivity, are ideal for flexible strain/pressure sensors, yet integrating high toughness, sensitivity, and conductivity remains challenging. This study fabricates a multifunctional hydrogel using methacrylated chitosan (CSMA) and acrylamide (AM) as the matrix, innovatively incorporating anhydrous betaine (Bet) and sodium chloride (NaCl). Bet remarkably enhances the compressive strength and mechanical properties, while dissociated NaCl provides abundant free ions, significantly boosting ionic conductivity. The resulting hydrogel demonstrates outstanding overall performance: a toughness of 6.99 MJ/m³, an ionic conductivity of 5.56 S/m, a compressive stress of 6.68 MPa at 90% strain, and a high self-healing efficiency of 96.6% within 16 h. Based on this hydrogel, the fabricated strain and pressure sensors exhibit high sensitivity (gauge factor GF = 3.18 within 250%–500%; pressure sensitivity $S_1 = 0.642 \text{ kPa}^{-1}$ within 0–50 kPa), enabling effective monitoring of diverse human motions. Furthermore, this work successfully extends the application of the hydrogel sensor to information transmission and handwritten character recognition, offering a novel strategy for designing high-performance flexible sensors.

1. Introduction

Flexible Conductive hydrogels with a three-dimensional network structure, boasting unique advantages of flexibility, self-healing ability and biocompatibility, could precisely meet the application requirements of flexible sensing devices. They had thus emerged as ideal materials in the field of intelligent sensor fabrication and garnered extensive attention and wide applications in relevant research [1–4].

In the selection of hydrogel substrates, natural polysaccharides such as cellulose, chitosan (CS) and sodium alginate have become preferred candidates for constructing high-performance hydrogels, owing to their excellent biocompatibility and environmental friendliness. Among them, chitosan, as a kind of alkaline polysaccharide in nature, is prepared by the deacetylation of chitin. The abundant amino and hydroxyl groups on its molecular chains endue the material with unique chemical activity, demonstrating enormous application potential in the field of conductive hydrogels [5–7]. However, traditional chitosan hydrogels generally suffer from inherent defects of poor toughness and weak compression resistance. This shortcoming greatly limits their large-scale

application in wearable flexible sensors.

In the field of chitosan hydrogel performance regulation, the introduction of salt components is a core strategy to optimize their network structure and functional properties. In previous studies, Sui et al. [8] have constructed a double-network system based on ammonium chloride-betaine/proline. The prepared composite hydrogel could maintain excellent antifreezing performance even at an extreme low temperature of -40°C , with an electrical conductivity of up to 2.72 S/m. Yang et al. [9] have utilized the coordination effect between betaine and metal ions (Al^{3+}) to prepare polyacrylic acid (PAA)-based composite hydrogels, which achieved a high conductivity of 6.74 S/m and a high breaking strength of 1.088 MPa, realizing the synchronous improvement of conductivity and mechanical properties. Betaine is a natural zwitterionic osmolyte derived from plants, which exhibits excellent biodegradability, chemical stability, and biocompatibility [10–12]. Its zwitterionic groups can strongly bind with water molecules, reducing the proportion of free water in the system, thereby inhibiting ice crystal nucleation and growth. Additionally, this substance can form flexible cross-linking bridges between chitosan molecular chains, significantly

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enhancing the toughness and tensile extensibility of hydrogels. However, its own contribution to increasing the system's conductivity is relatively weak. Inorganic salts, upon dissociation, can effectively improve the conductive ability of hydrogels [13–15]; there has been some research accumulation in this area. Nevertheless, systematic investigations into the synergistic effects of inorganic salts and zwitterions, as well as the regulatory mechanisms governing the mechanical and conductive properties of hydrogels, remain lacking.

To address the above issues, this study proposes an innovative strategy that employs CSMA and AM as polymer monomers and introduces a Bet-NaCl composite system to fabricate chitosan-based conductive hydrogels (Fig. 1). By exploiting the ionic coordination between Na^+ and the carboxylate groups ($-\text{COO}^-$) of betaine, together with the ion-pair association between Cl^- and the quaternary ammonium groups ($-\text{N}(\text{CH}_3)_3^+$) of betaine, this strategy simultaneously achieves mechanical reinforcement of the hydrogel network and the construction of efficient ion transport pathways at the molecular level, thereby overcoming the long-standing trade-off between electrical and mechanical properties in conventional conductive hydrogels. As a result, the hydrogel delivers a high compressive strength of 6.68 MPa, a high toughness of 6.99 MJ/m^3 , and a high ionic conductivity of 5.56 S/m . Furthermore, X-ray photoelectron spectroscopy (XPS) analysis provides direct experimental evidence between inorganic salts and zwitterions. Moreover, the flexible strain/pressure sensor fabricated from this hydrogel integrates a wide sensing range (strain: 0–500%, pressure: 0–2000 kPa), high sensitivity ($\text{GF} = 3.18$, $S_1 = 0.642 \text{ kPa}^{-1}$), excellent self-healing capability (96.6% healing efficiency within 16 h), and anti-freezing performance (retaining an ionic conductivity of 3.33 S/m and mechanical flexibility at -20°C). These combined properties render the sensor highly promising for applications in human motion monitoring, physiological signal acquisition, encrypted information transmission, and human-machine interaction, providing a new material platform and technical pathway for the design of next-generation green, intelligent flexible wearable electronics.

2. Materials and methods

2.1. Materials

Chitosan (CS, degree of deacetylation: 90%, $\text{Mw} = 200,000 \text{ Da}$) was supplied by Shanghai Macklin Biochemical Technology Co., Ltd. (China); Methacrylic anhydride (MA, 98.04%) was purchased from Pisco; Acetic acid (HAc) was obtained from Hengxing; Acrylamide (AM, $\geq 98\%$) and Betaine (Bet, 98%+) were purchased from Adamas; Sodium chloride (NaCl , $\geq 99.9\%$) was bought from Shanghai Titan Technology Co., Ltd.; 2-hydroxy-4'-(2-hydroxyethoxy)-2-methylpropiophenone (Photoinitiator 2959, 99%) was purchased from Shanghai Bide Pharmaceutical Technology Co., Ltd. (Shanghai, China); and deionized water was self-prepared in the laboratory.

2.2. Preparation of the CSMA-AM@Bet-NaCl hydrogel

First, the prepared CSMA, 2 wt% was dispersed in deionized water at room temperature, and vigorous stirring was maintained until a uniform and transparent solution was formed. Subsequently, 0.20 g of AM, 0.05 g of NaCl , 0.02 g of Bet, and 0.005 g of initiator 2959 were added to the aforementioned mixture. The solution was stirred at room temperature to form a precursor solution. Then, the precursor solution was poured into a mold, and photopolymerization was carried out under ultraviolet (UV) light (40 W) for 30 s.

2.3. Characterizations

The CSMA sample was dissolved in deuterated reagent heavy water (D_2O) respectively and adopted Bruker AVANCE III NMR hydrogen spectroscopy (500 MHz). The microstructure of the gel was observed by Quattro S field emission scanning electron microscope (ESEM) The FT-IR data of dry CAB hydrogels were tested by a Shimadzu IR PRESTIGE-21 spectrometer with the KBr tablet method. Using a controlled-stress rheometer (MCR302 SN80987260), we examined the rheological characteristics of CSMA-AM@Bet-NaCl gels. All the stress-

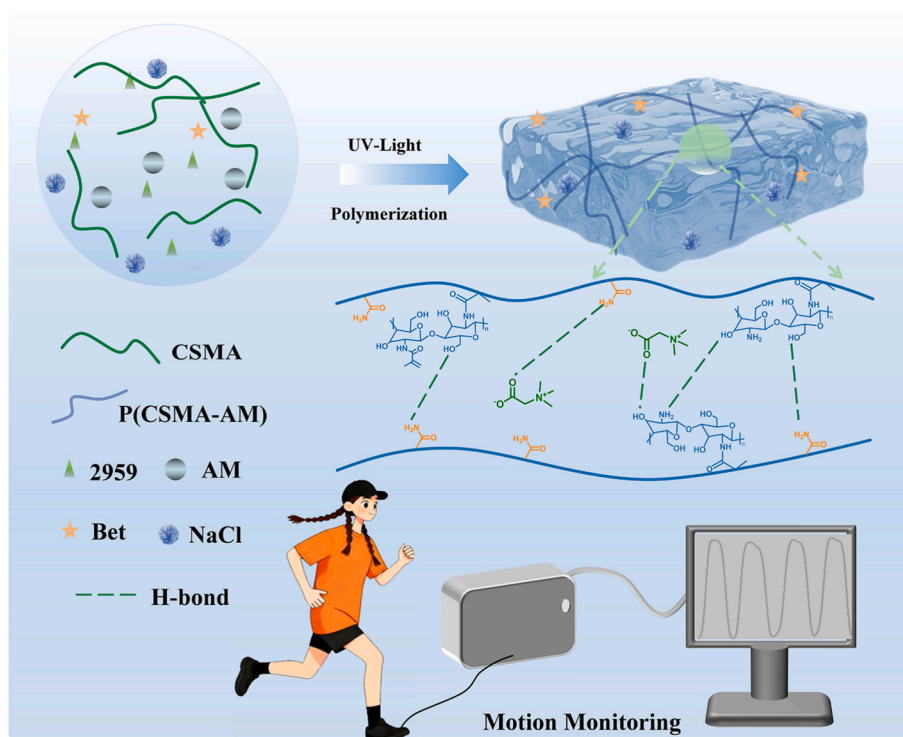


Fig. 1. Schematic representation of the preparation procedure and cross-linking structure of (CSMA-AM)@Bet-NaCl hydrogel.

strain curves of the CSMA-AM@Bet-NaCl hydrogel strips ($2\text{ cm} \times 2\text{ cm} \times 0.2\text{ cm}$) were examined using a microcomputer-controlled electronic universal testing machine (IGDX2-60 + 300B1, Lishi Instruments Co., Ltd.) at a stretching speed of 100 mm/min and an actual gripping distance of 2 cm (Shanghai, China). The mixed hydrogel precursors were poured into the mold with the LED ultraviolet light source of LWGGJ-LED under an ultraviolet lamp with 40 W power for gel formation.

2.4. Mechanical property testing

Square hydrogel samples (10 mm in diameter and 5 mm in height) were prepared for compression testing. The experiment was conducted at compression ratios of 20%, 40%, 60%, 80%, and 90%, respectively, with a compression rate of 100 mm/min. The compressive mechanical properties and cyclic stability of the hydrogels were analyzed.

2.5. Conductivity test

The electrical conductivity of the hydrogels was investigated via electrochemical impedance spectroscopy (EIS). Impedance data were collected using an electrochemical workstation (CHI760E, Shanghai Chenhua Instrument Co., Ltd., China). The conductivity (σ) of the hydrogels was calculated using the formula $\sigma = L/(R \times S)$, where S (cm^2) and R (Ω) represent the cross-sectional area and resistance of the hydrogel, respectively, and L (cm) denotes the distance between the two probes.

2.6. Adhesion test

Hydrogel samples with dimensions of $10\text{ mm} \times 10\text{ mm} \times 1\text{ mm}$ (length $\times$ width $\times$ thickness) were sandwiched between two rectangular substrate sheets, with an adhesion area of $10\text{ mm} \times 10\text{ mm}$. The tensile rate was set at 100 mm/min.

2.7. Self-healing and testing

The self-healing efficiency of the hydrogels was evaluated by comparing the tensile tests of the original samples and the healed samples. The healing efficiency (HE%) was calculated using the equation: $\text{HE}(\%) = \text{Sh} / \text{S}_0 \times 100\%$, where S_0 is the tensile strength of the original sample, and Sh is the tensile strength of the self-healed sample.

2.8. Strain sensing

To monitor human movement, the hydrogels were attached to different parts of the human body. The relative resistance change ($\Delta R/R_0$) was derived from the equation $\Delta R/R_0 = (R - R_0)/R_0 \times 100\%$, where R_0 represents the initial resistance, and R denotes the resistance of the hydrogel under strain. The gauge factor (GF) was used to evaluate the sensitivity of the hydrogels, which is defined as the ratio of the relative resistance change ($\Delta R/R_0$) to the applied strain (ϵ): $\text{GF} = (\Delta R/R_0)/\epsilon$.

2.9. Pressure sensing

The relative current change ($\Delta I/I_0$) was obtained using the equation $\Delta I/I_0 = (I - I_0) / I_0 \times 100\%$, where I_0 represents the initial current, and I denotes the current of the hydrogel under pressure. The pressure sensitivity (S) of a single device was defined as $S = (\Delta I/I_0)/\Delta P$, where ΔI is the current change, I_0 is the initial current, I is the current under applied pressure, and ΔP represents the change in applied pressure.

2.10. Synthesis of CSMA

CS was first dissolved to prepare a chitosan solution prior to the grafting reaction. Specifically, 1 g of CS was added into 100 mL of 1% (v/v) acetic acid solution at room temperature, and the mixture was stirred

thoroughly until CS was completely dissolved, resulting in a 1% (w/v) chitosan solution. Subsequently, 2 mL of MA was added dropwise into the above chitosan solution under stirring. After the completion of MA addition, the reaction mixture was transferred to a water bath set at 60°C , and the grafting reaction was allowed to proceed for 24 h under constant stirring to ensure sufficient reaction between CS and MA.

2.11. Biocompatibility testing

The CSMA-AM@Bet-NaCl gel was soaked in culture medium at a ratio of $2\text{ cm}^2/\text{mL}$ and extracted at 37°C for 24 h. After extraction, the medium was centrifuged at 1000 rpm for 5 min, and the supernatant was collected to obtain the gel extract with a concentration of $2\text{ cm}^2/\text{mL}$. It was further diluted with culture medium to prepare extracts at concentrations of $0.1\text{ cm}^2/\text{mL}$ and $0.5\text{ cm}^2/\text{mL}$ for experimental groups.

The gel extracts were added into 96-well plates. L-929 cells were seeded into each well, and 100 μL of sterile phosphate-buffered saline (PBS) was supplemented. The plates were then incubated in a humidified incubator with 5% CO_2 atmosphere at 37°C for 1, 2 and 3 days, respectively.

2.12. Cell proliferation assay

Cell proliferation was evaluated using the Cell Counting Kit-8 (CCK-8) assay. The 96-well plate was removed from the incubator, and 10 μL of DMEM-CCK-8 mixture was added to each well. The plate was then incubated for an additional 3 h at 37°C in the incubator. The optical density (OD) value was measured at 450 nm using a microplate reader to assess cell viability.

3. Results and discussion

3.1. Design and characterization of CSMA-AM@Bet-NaCl

Fig. 1 illustrates the synthesis route of the CSMA-AM@Bet-NaCl hydrogel. CSMA and AM were employed as comonomers, which constructed the hydrogel matrix via intermolecular hydrogen bonding interactions. The Bet-NaCl composite system was incorporated, where Bet significantly enhanced the maximum compressive pressure of the hydrogel to optimize its mechanical properties, while dissociated NaCl provided abundant free ions to effectively improve the ionic conductivity of the hydrogel. Ultimately, a hydrogel material with both excellent mechanical properties and electrical conductivity was successfully fabricated.

CSMA and CSMA-AM@Bet-NaCl hydrogels were characterized by Fourier-transform infrared spectroscopy (FT-IR), proton nuclear magnetic resonance spectroscopy (^1H NMR), energy-dispersive X-ray spectroscopy (EDS), X-ray diffraction (XRD), and X-ray photoelectron spectroscopy (XPS). Meanwhile, environmental scanning electron microscopy (ESEM) characterization and rheological tests of the hydrogels were also performed. Relevant results and corresponding images (Fig. S1-S2) are presented in the Supporting Information.

3.2. Mechanical property

The hydrogel exhibited excellent flexibility, and its microstructure remained intact after undergoing deformations such as stretching, twisting, and knotting under external forces. Moreover, it achieved rapid rebound upon the removal of compressive external forces, which fully demonstrated its outstanding mechanical properties (Fig. S3). Meanwhile, even at -20°C , the hydrogel can still withstand bending, knotting, and stretching while maintaining its structural integrity (Fig. S4a). Hydrogels without added Bet or NaCl can still be bent and tied at -20°C , but their gel matrix turns white, and their freeze resistance is significantly weaker than that of CSMA-AM@Bet-NaCl hydrogels. At low temperatures, these hydrogels become brittle and cannot sustain tensile

deformation (Fig. S4b-c). DSC test results indicate that CSMA-AM@NaCl and CSMA-AM@Bet hydrogels have lower low-temperature phase transition temperatures. The CSMA-AM@Bet-NaCl hydrogel exhibits a distinct endothermic peak at -21°C , demonstrating that this material can effectively inhibit ice crystal formation. Bet and NaCl exhibit a synergistic effect in lowering the freezing point, endowing the hydrogel with excellent antifreeze properties (Fig. S4d). The zwitterionic groups of betaine ($-\text{COO}^-$ and $-\text{N}^+(\text{CH}_3)_3$) are tightly bound to water molecules through electrostatic interactions and hydrogen bonding, significantly reducing the content of free water and effectively inhibiting ice crystal growth. The Na^+ and Cl^- ions dissociated from NaCl further enhance the gel's hydration through ion-dipole interactions, thereby lowering the freezing point.

As shown in Fig. S5 and Table S1, 2 wt% CSMA and 0.20 g AM increased the polymer content within the gel matrix, thereby facilitating the formation of hydrogen bonds and chain entanglements between polymer chains, and the hydrogel presented superior comprehensive mechanical properties [16]. After the introduction of Bet into the system (Fig. S6 and Table S2), the $-\text{N}^+(\text{CH}_3)_3$ and $-\text{COO}^-$ groups increased the number of dynamic cross-linking points inside the hydrogel, and its mechanical properties were further improved. However, the electrical conductivity of the hydrogel remained at a relatively low level at this stage. To address this issue and enhance the conductivity of the hydrogel, NaCl was incorporated into the system. Fig. S7 and Table S3 displayed the stress-strain curves of the hydrogels with different NaCl addition amounts. Figs. 2a-b show the effect of different components on the tensile properties of the gel. When the system formulation was fixed as 0.6 mL of 2 wt% CSMA, 0.20 g of AM, 0.05 g of NaCl and 0.02 g of Bet, the hydrogel achieved optimal mechanical properties, with a toughness of 6.99 MJ/m^3 and a Young's modulus of 493 kPa.

As illustrated in Fig. 2c and d, with the strain increasing from 100% to 300%, the calculated dissipation rate rose from 23.67% to 28.80%. The main source of dissipated energy was the breakage and reconstruction of hydrogen bonds inside the gel network. Due to the inability of hydrogen bonding interactions to recover rapidly and completely, the material exhibited significant hysteresis behavior [17,18]. To further evaluate the mechanical stability and elastic recovery performance of the material, ten consecutive loading-unloading cycles were carried out at strains of 50%, 100%, 150%, 200%, 250% and 300% (Fig. S8a-f). The high overlap of tensile loading and unloading curves reveals that the hydrogel possesses superior mechanical recoverability during stretching. Furthermore, the dissipation ratio was calculated according to ten cyclic curves at 100% strain (Fig. S8h). It decreased from 32.68% in the first cycle to 24.59% in the second cycle and then remained stable, suggesting that the hydrogel can quickly reach dynamic equilibrium upon repeated stretching, and no obvious irreversible damage occurs to its internal network structure. Besides, 100 loading-unloading cycles were tested under a constant strain of 100% (Fig. S8g). The hysteresis loop area slightly reduces and gradually stabilizes with the increase of cycle numbers [19,20]. In addition, we performed 1000 stretching cycles on the CSMA-AM@Bet-NaCl hydrogel at 100% strain. As shown in Fig. S9, the sensor maintained a stable electrical signal output throughout the cycling process, fully demonstrating its excellent durability and long-term operational stability as a flexible strain sensor.

The compressive mechanical properties of the hydrogel were further evaluated via uniaxial compression tests and cyclic loading-unloading compression tests. Its compressive properties were jointly regulated by the CSMA concentration and AM content (Fig. S10). At a CSMA concentration of 2 wt%, a uniform and dense cross-linked network was formed, and the maximum compressive stress of the system consequently reached a peak value. At 3 wt% CSMA, the excessive CSMA led to enhanced rigidity of the network structure and restricted the movement of molecular chain segments, resulting in an opposite decrease in the maximum compressive stress [21,22]. As shown in Fig. 2e, when the compressive strain increased from 20% to 90%, the maximum compressive strength could reach 0.11 MPa to 6.68 MPa, and no obvious

structural rupture occurred during the deformation process, which fully proved that it possessed excellent structural stability and compression resistance. For generating reliable and stable output signals, the mechanical stability of the hydrogel pressure sensor was essential. As presented in Fig. 2f and g, the fatigue resistance of the hydrogel was evaluated via cyclic compression loading-unloading tests with 10 consecutive compression cycles at 10% strain. Similar to the results of tensile cycles, after an initial decline, the calculated dissipation rate decreased from 10.41% to 10.7% and then remained basically unchanged, indicating that the hydrogel had remarkable fatigue resistance and self-recovery capability.

The compression performance of chitosan-acrylamide (CSMA-AM)-based hydrogels was systematically characterized and analyzed through compression stress-strain curves (Figs. 2h-i, S11, and Table S4). The core mechanism for enhancing mechanical properties lies in the introduction of betaine: betaine molecules can act as hydrogen bond mediators, inducing the formation of a denser and more stable crosslinked network between CSMA-AM molecular chains. This high-density hydrogen bond structure effectively hinders the sliding, deformation, and dissociation of molecular chains under external compression, significantly improving the hydrogel's compressive strength and structural stability [23,24]. Under 90% compressive strain conditions, the compression strength of CSMA-AM@Bet-NaCl hydrogels reaches 6.68 MPa, which is three times that of CSMA-AM@NaCl hydrogels without betaine addition (2.23 MPa), directly confirming the enhancing effect of betaine on mechanical properties. The improvement in conductivity stems from the action of sodium chloride in the system: sodium chloride dissociates into freely mobile Na^+ and Cl^- ions within the hydrogel pores. These ions can rapidly migrate within the gel network, constructing efficient ionic conduction pathways, thereby significantly enhancing the material's conductivity [25–27]. Fig. 2j shows the electrochemical impedance spectra of hydrogels with different mass ratios of Bet to NaCl. The hydrogel with a Bet:NaCl ratio of 2:5 exhibits the lowest impedance value and the highest ionic conductivity on the EIS curves. As shown in Fig. 2k and l, there is a metal coordination interaction between Na^+ and betaine, which exhibits excellent tensile strength and maximum fracture strain, achieving an optimal balance between electrical performance and mechanical flexibility. Therefore, Bet: NaCl = 2:5 was selected as the optimal ratio in this study, realizing the synergistic regulation of the mechanical and electrochemical properties of the hydrogel. In summary, through betaine-mediated hydrogen bond crosslinking enhancement and ion conduction pathways constructed by sodium chloride, this hydrogel achieves synergistic improvement in mechanical properties and conductivity. This lays a solid foundation for its application in strain/pressure sensor fields and demonstrates broad development prospects.

3.3. Self-healing properties

Flexible sensors are prone to damages such as scratches and punctures during service, which can degrade their stability and shorten their service life. The self-healing capability enables flexible sensors to recover their sensing performance after damage, thereby avoiding the issues of poor stability and reduced service life caused by damage. A comparison of the hydrogel's resistance before and after healing revealed no significant difference between the two (Fig. S12), further verifying that the CSMA-AM@Bet-NaCl hydrogel not only exhibits a fast self-healing rate but also can well restore its initial conductive performance after healing. This provides critical performance support for its long-term cyclic application in the field of flexible electronics.

Mechanistically, when the hydrogel was cut, the fractured surfaces could achieve self-healing through the reformation of dynamic hydrogen bonds. Notably, the healed hydrogel still retained excellent flexibility and could normally withstand deformations such as bending and twisting (Fig. 3a). The self-healing process of the hydrogel was further observed under an optical microscope: the internal structure of

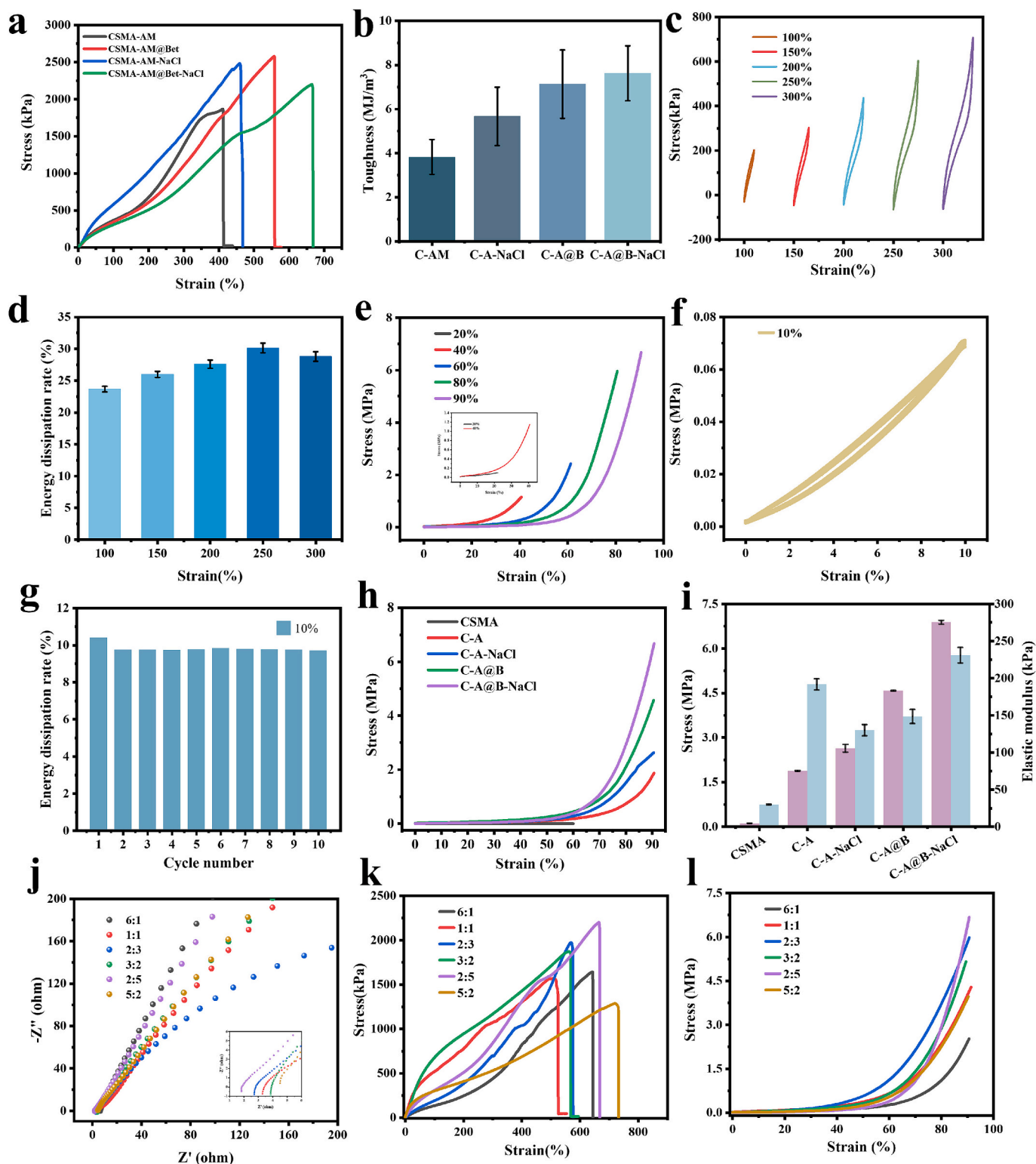


Fig. 2. (a) Tensile stress-strain curves of hydrogels with different components (CSMA-AM, CSMA-AM@Bet, CSMA-AM@NaCl, CSMA-AM@Bet-NaCl); (b) Corresponding toughness of hydrogels with different components; (c) Tensile stress-strain curves of CSMA-AM@Bet-NaCl hydrogel at different elongations (100%, 150%, 200%, 250%, 300%); (d) Energy dissipation of CSMA-AM@Bet-NaCl hydrogel at different elongations; (e) Compressive stress-strain curves of CSMA-AM@Bet-NaCl hydrogel at different compression ratios (20%, 40%, 60%, 80%, 90%); (f) Stress-strain curves of CSMA-AM@Bet-NaCl hydrogel during ten loading-unloading cycles at 10% compression; (g) Energy dissipation of CSMA-AM@Bet-NaCl hydrogel during ten loading-unloading cycles at 10% compression; (h) Compressive stress-strain curves of CSMA, CSMA-AM, CSMA-AM@NaCl, CSMA-AM@Bet, and CSMA-AM@Bet-NaCl hydrogels; (i) Corresponding compressive strength and Young's modulus of CSMA, CSMA-AM, CSMA-AM@NaCl, CSMA-AM@Bet, and CSMA-AM@Bet-NaCl hydrogels; (j) EIS plots of hydrogels with different Bet/NaCl ratios (6:1, 1:1, 2:3, 3:2, 2:5, 5:2); (k) Tensile stress-strain curves of hydrogels with different Bet/NaCl ratios; (l) Compressive stress-strain curves of hydrogels with different Bet/NaCl ratios.

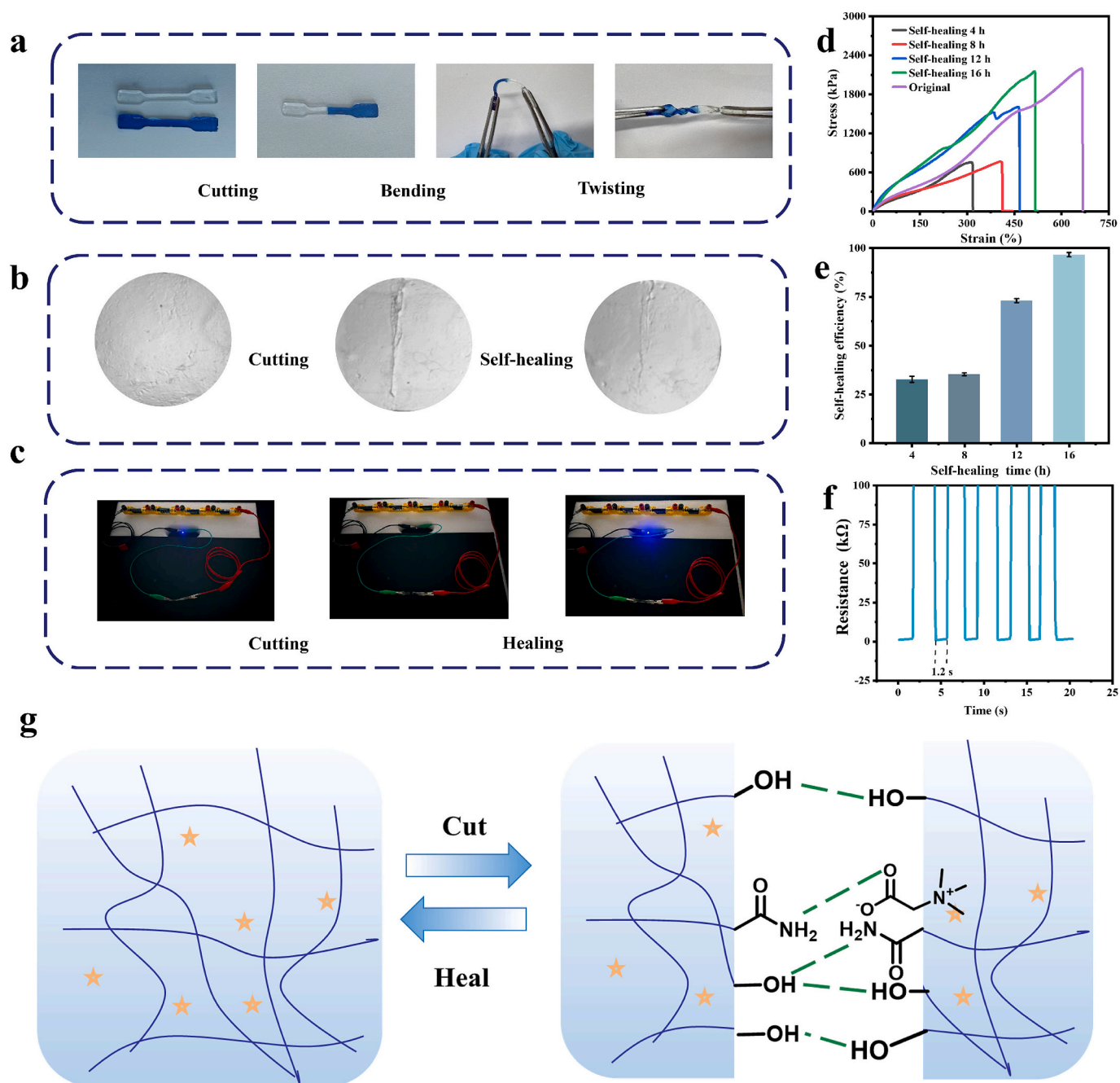


Fig. 3. (a) Macroscopic images of the self-healing behavior of hydrogels; (b) Microscopic images of the self-healing process of hydrogels; (c) Brightness variation of blue LED before and after hydrogel self-healing; (d) Mechanical self-healing properties of hydrogels; (e) Mechanical self-healing efficiency of hydrogels at different healing times; (f) Resistance variation during the electrical self-healing process; (g) Schematic diagram of the self-healing mechanism of CSMA-AM@Bet-NaCl hydrogel.

the hydrogel was almost completely healed within 2 h, with nearly no visible traces of the cut (Fig. 3b). To verify the self-healing-induced conductivity recovery, the hydrogel was integrated into a closed circuit to illuminate a blue light-emitting diode (LED). Owing to the hydrogel's excellent electrical conductivity, the blue LED was reilluminated when the cut hydrogel segments were brought into contact again (Fig. 3c), demonstrating that the hydrogel maintained favorable conductivity and met the application requirements of flexible wearable electronic devices [28].

To evaluate the self-healing capability of the hydrogel's dynamic network, tensile property tests were conducted on hydrogel samples with different self-healing durations (Fig. 3d). The experimental results

showed that the stress recovery degree of the hydrogel gradually increased with the extension of self-healing time; when the self-healing time reached 16 h, the stress self-healing efficiency could be as high as 96.6% (Fig. 3e). These results fully confirm that the hydrogel possesses excellent self-healing efficiency and repair effects, providing reliable material support for improving the durability of flexible sensors. The electrical recovery capability of the hydrogel was further evaluated by monitoring the resistance variation during consecutive cutting-self-healing cycles (Fig. 3f). After the hydrogel was sheared, its resistance increased significantly; however, once the hydrogel fragments were in contact, the resistance recovered to its initial value within 1.2 s, confirming its ultra-high healing efficiency. The excellent self-healing

capability arises from the synergy of multiple reversible non-covalent interactions. Abundant dynamic hydrogen bonds form among hydroxyl, amino, amide, carboxylate, and quaternary ammonium groups. Driven by these combined mechanisms, the material can autonomously heal cracks at room temperature without any external stimulus (Fig. 3g).

3.4. Adhesive properties

Fig. S13a illustrates the adhesion mechanism of the hydrogel. Owing to the presence of abundant polar groups ($-C=O$, $-OH$, $-NH_2$) on the hydrogel chains, hydrogen bonds were formed between these groups and the functional groups ($-NH_2$ and $-OH$) on the substrate surfaces [29,30]. This interaction enabled the hydrogel to adhere firmly to various substrates, including cardboard, plastic, wood, glass, balloons, and skin (Fig. S13b) [31,32]. To quantify its adhesion capability, lap shear tests were conducted to determine the adhesion strength of the hydrogel on different substrate surfaces (Fig. S13c). The adhesion strength-displacement curves of the hydrogel are presented in Fig. S13d; specifically, the adhesion strength of the hydrogel to pig skin reached 61.9 kPa (Fig. S13e), demonstrating its ability to effectively adhere to human skin. As shown in Fig. S13f, the hydrogel could still maintain adhesion to cardboard even after ten cycles of attachment and detachment, indicating excellent reusability. After ten cycles, the lap shear adhesion strength of the hydrogel was maintained at 202.1 kPa, corresponding to a retention rate of 74% relative to the initial adhesion strength of 271.6 kPa. Collectively, these results confirm that the hydrogel exhibits outstanding adhesion performance, making it a promising candidate material for flexible sensors.

3.5. Electrochemical performance and strain sensing performance

High electrical conductivity is conducive to improving the sensing performance of sensors; thus, the effect of NaCl addition on the conductive properties was investigated. As shown in Fig. 4a, the prepared hydrogel was introduced into the circuit as a conductor. With the gradual stretching of the hydrogel, the ion transport paths were compressed, leading to a sharp increase in resistance and a gradual dimming of the light-emitting diode (LED) lamp [33–35]. The electrical conductivity of the CSMA-AM@Bet hydrogel was only 0.091 S/m (Figs. S13a–b). As the NaCl content increased from 0.01 g to 0.05 g, the electrical conductivity of the hydrogel continuously increased from 1.39 S/m to 5.56 S/m (Figs. 4b–c), indicating that the introduction of NaCl exerted a significant regulatory effect on the hydrogel's electrical conductivity [36]. When the amount of NaCl added increases to 60 mg, the system exhibits supersaturation and salting-out effects. Excess salt ions precipitate and hinder ion migration, leading to a decrease in the hydrogel's conductivity.

Additionally, the impedance of the hydrogel decreased with increasing temperature, which implied enhanced conductivity, reaching 7.81 S/m at 80 °C (Fig. 4d). This increase might be related to the enhanced movement of polymer segments and improved ion migration with rising temperature. Furthermore, the introduction of NaCl further reduced the freezing point of the hydrogel, endowing it with excellent antifreeze performance [37,38]. At -20 °C, the electrical conductivity of the hydrogel still reached 3.33 S/m (Fig. 4e). Sensitivity is the most critical factor for evaluating sensing performance, and the gauge factor (GF) serves as an indicator to measure the sensitivity of sensors [39,40]. With the increase in applied tensile strain, the resistance change gradually increased. In the strain range of 0–250%, the linear fitting coefficient $R^2 = 0.96$ with a gauge factor (GF) of 1.48; in the range of 250%–500%, $R^2 = 0.99$ and $GF = 3.18$ (Fig. 4f). These results demonstrate that the hydrogel sensor exhibits excellent linear response characteristics within each defined strain interval, ensuring the accuracy of signal output for practical applications. Moreover, as shown in Fig. 4g, the signal output gradually increased and remained stable as the strain rose from 0% to 400%.

As illustrated in Figs. 4h–i, the hydrogel sensor could stably and rapidly respond to a wide strain range, from small strains of 0.1% to large strains of 300%. Subsequently, tests were conducted to evaluate the fast response performance of the hydrogel sensor. It can be seen from Fig. 4j that both the response time and recovery time of the sensor were 0.3 s, fully demonstrating its excellent dynamic response capability [41]. The strain sensing curves of the hydrogel under different stretching speeds are presented in Fig. 4k; when the stretching speed increased from 50 mm/min to 400 mm/min, the sensing curves still exhibited stable strain sensing performance. After self-healing treatment of the hydrogel, the output level of its sensing signal showed little difference from the original state before damage, enabling effective recovery of sensing performance (Fig. 4l). The aforementioned experimental results indicated that the hydrogel possessed excellent strain response characteristics, holding potential application prospects in high-performance strain sensors.

3.6. Pressure sensing performance

Owing to its high electrical conductivity and compressibility, the hydrogel could also serve as an ideal material for developing flexible pressure sensors. Fig. 5a presents the compression schematic diagram of the hydrogel. Fig. 5b shows the relationship between the relative current change ($\Delta I/I_0$) and pressure (0–2000 kPa). The pressure sensing range of this hydrogel could be divided into three intervals: 0–50 kPa, 50–500 kPa, and 500–2000 kPa, with corresponding sensitivities (S) of $S_1 = 0.642 \text{ kPa}^{-1}$, $S_2 = 0.127 \text{ kPa}^{-1}$, and $S_3 = 0.047 \text{ kPa}^{-1}$, respectively. This indicated that the hydrogel pressure sensor possessed a wide sensing range and high pressure sensitivity. The sensitivity decreases gradually with rising applied pressure. This can be attributed to the stepwise structural evolution of the hydrogel network under increasing compression. In the initial compression region of 0–50 kPa, even a slight pressure increment can greatly enlarge the actual contact area, constructing abundant ion transport pathways at the interface and achieving the highest sensitivity of 0.642 kPa^{-1} . As pressure further increases, the sensing mechanism shifts to the elastic deformation of the bulk porous network. Within the range of 50–500 kPa, owing to the elastic restoring force of the dual-network structure, a larger pressure increment is required to achieve equivalent further deformation, thus resulting in lower sensitivity compared with the 0–50 kPa region. Under high-pressure compression of 500–2000 kilopascals, polymer chains are squeezed, which greatly hinders further structural deformation. Consequently, a sharp increase in pressure only triggers a trivial change in relative resistance ($\Delta R/R_0$), leading to the lowest sensitivity in this pressure region [42–44]. As illustrated in Fig. 5c–d, the current responses of the pressure sensor were tested under low pressures (0.5 kPa, 1 kPa, 5 kPa, 10–50 kPa) and high pressures (100–500 kPa, 1.0 MPa, 1.5 MPa, and 2.0 MPa). The results showed that the hydrogel sensor could accurately reflect the magnitude of the applied force.

In addition, the current-voltage (I - V) curves of the hydrogel pressure sensor under different pressure values were measured (Fig. 5e), and these I - V curves exhibited excellent sensitivity to pressures ranging from 1 to 10 kPa. When the hydrogel sensor was subjected to a strain of 50 kPa, it maintained stable ($\Delta I/I_0$) signals at different stretching rates (10–30 N/S) (Fig. 5f). As shown in Fig. 5g, the resistance of the sensor increased accordingly after a load was applied to its surface; once the load was removed, the resistance rapidly recovered to its original state, indicating that the sensor exhibited a stable response under various loads.

Furthermore, when the sensor was used to monitor different motion states (walking and running), the displayed signals differed in peak amplitude (Fig. 5h). Additionally, the hydrogel demonstrated a fast response-recovery time under a pressure of 50 kPa (Fig. 5i), which provided crucial support for its application in long-term monitoring scenarios of flexible wearable devices. As shown in Fig. S15a, the resistance variation of the sensor during the mouse-clicking cycle

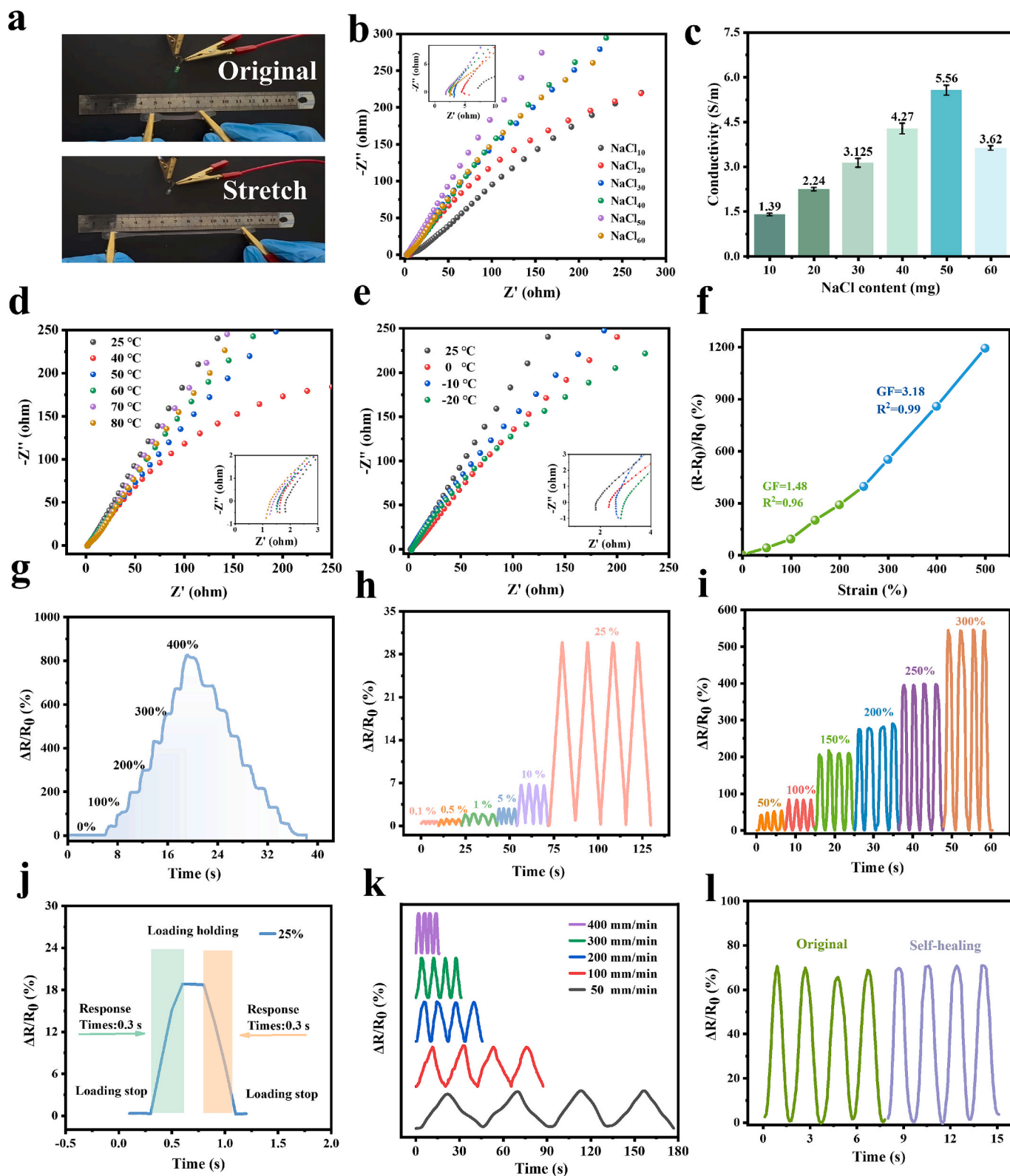


Fig. 4. Strain-sensing performance of the hydrogel. (a) Brightness change of small light bulb when stretching the hydrogel. (b-c) Electrochemical impedance spectra (EIS) and corresponding electrical conductivities of hydrogels with different NaCl contents. (d) Impedance plots of CSMA-AM@Bet-NaCl hydrogel at high temperatures (25–80 °C). (e) Impedance plots of CSMA-AM@Bet-NaCl hydrogel at low temperatures (–20–25 °C). (f) Relative resistance changes and GF of the strain sensor under different tensile strains. (g) $\Delta R/R_0$ of the hydrogel at strains ranging from 0% to 400%. (h-i) Relative resistance changes of the strain sensor under (h) small strains (0.1%–25%) and (i) large strains (50%–300%). (j) Response and recovery times of the hydrogel strain sensor. (k) Relative resistance changes of the hydrogel sensor at different tensile rates under 100% strain. (l) Sensing signals of the hydrogel before and after self-healing under 100% strain.

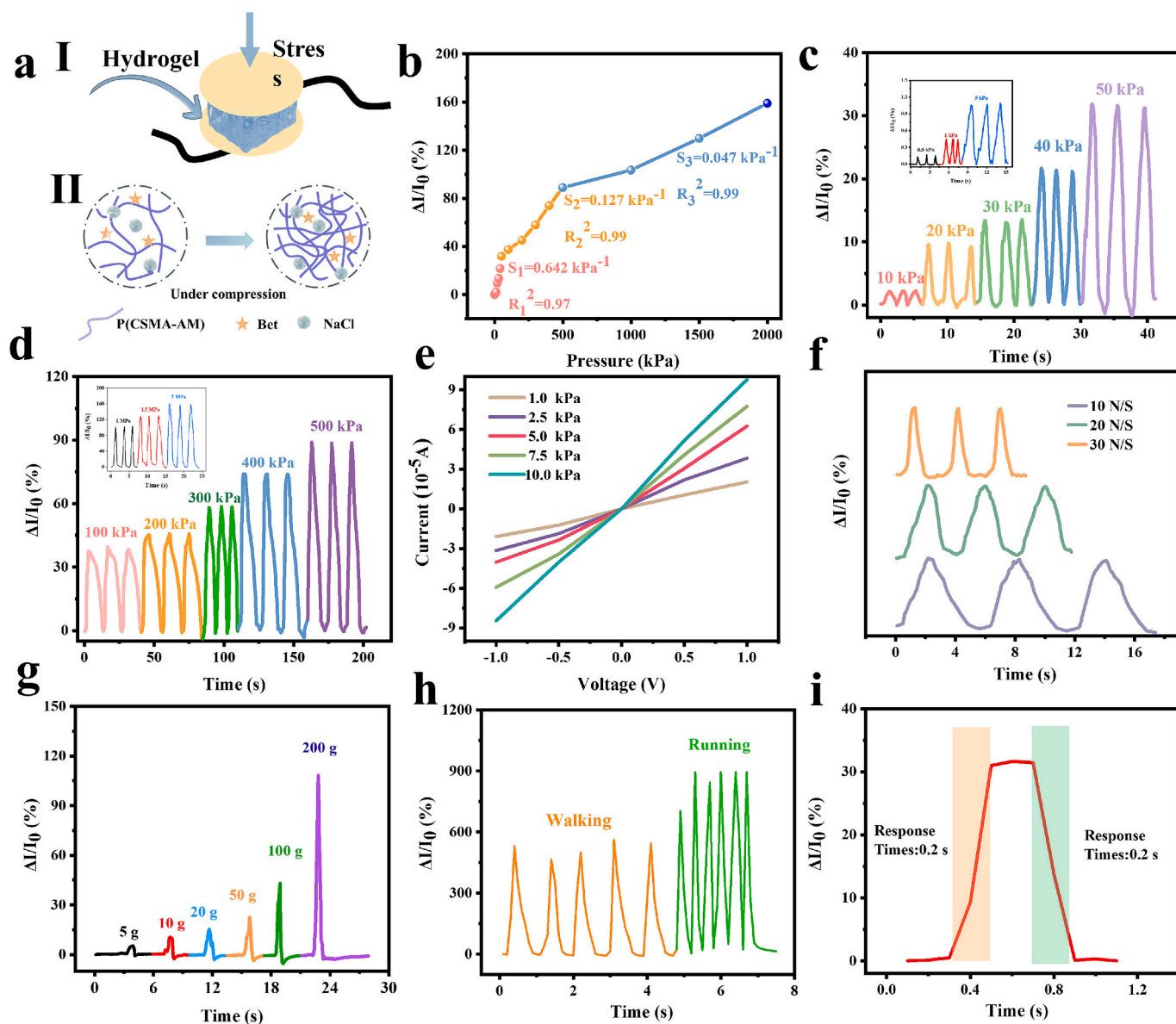


Fig. 5. Pressure-sensing performance of the hydrogel. (a) I Schematic diagram of compression sensing of hydrogel; II Schematic illustration of internal network structure of hydrogel under compression; (b) Sensitivity of the hydrogel pressure sensor in the range of 0–2000 kPa. (c) Lower detection limit of the hydrogel pressure sensor. (d) Upper detection range of the hydrogel pressure sensor. (e) I-V curves of the hydrogel as pressure increases from 1 kPa to 10 kPa. (f) Curves of the hydrogel under different compression rates at a pressure of 10 kPa. (g) Current changes under different weight loads; (h) Current output signals during walking and running. (i) Compressive response time of the hydrogel under a pressure of 50 kPa.

exhibited excellent repeatability, which indicated its outstanding sensing performance and cyclic stability. The balloon-pinching test demonstrated that when the sensor was attached to the balloon surface, it could efficiently respond to the dynamic changes in gas volume, suggesting its potential for further application in respiratory status monitoring (Figs. S15b-c). In addition, the hydrogel used in the sensor possessed both favorable mechanical flexibility and sensing properties. It could conform to the human body to detect swallowing movements (Fig. S15d) and maintain functionality even after multiangle twisting (Fig. S15e). Furthermore, the sensor achieved voice recognition by perceiving signal variations induced by vocal cord vibration, exhibiting excellent bioresponsive performance (Fig. S16). The cytocompatibility of the hydrogel was evaluated by the CCK-8 assay using L-929 fibroblasts, with the results shown in Fig. S17. At the low extract concentration of $0.1 \text{ cm}^2/\text{mL}$, cell viability remained above 80% even after 3 days of culture, indicating no obvious cytotoxicity. Although viability decreased with increasing extract concentration ($0.5 \text{ cm}^2/\text{mL}$) and

prolonged incubation, it still exceeded 60% on day 3, satisfying the basic biocompatibility requirements for biomaterials. These results confirm that the CSMA-AM@Bet-NaCl hydrogel possesses favorable cytocompatibility, supporting its potential use in wearable electronic devices.

Compared with hydrogels reported in the literature, the hydrogel strain sensor prepared in this study exhibited superior flexibility, a lower detection limit, and higher conductivity (Table S5 and S6), making it an ideal candidate for detecting signals related to human motion and physiological status. As shown in Fig. 6, the hydrogel strain sensor demonstrated stable performance in the detection of movements at body parts including the fingers, wrists, elbows, knees, neck, and face. When attached to the surface of a finger, the sensor could accurately detect the changes in the joint angle by sensing the deformation caused by finger joint movements (Fig. 6a).

Further test results indicated that the sensor could not only generate continuous responses to the dynamic movements of the wrist, elbow,

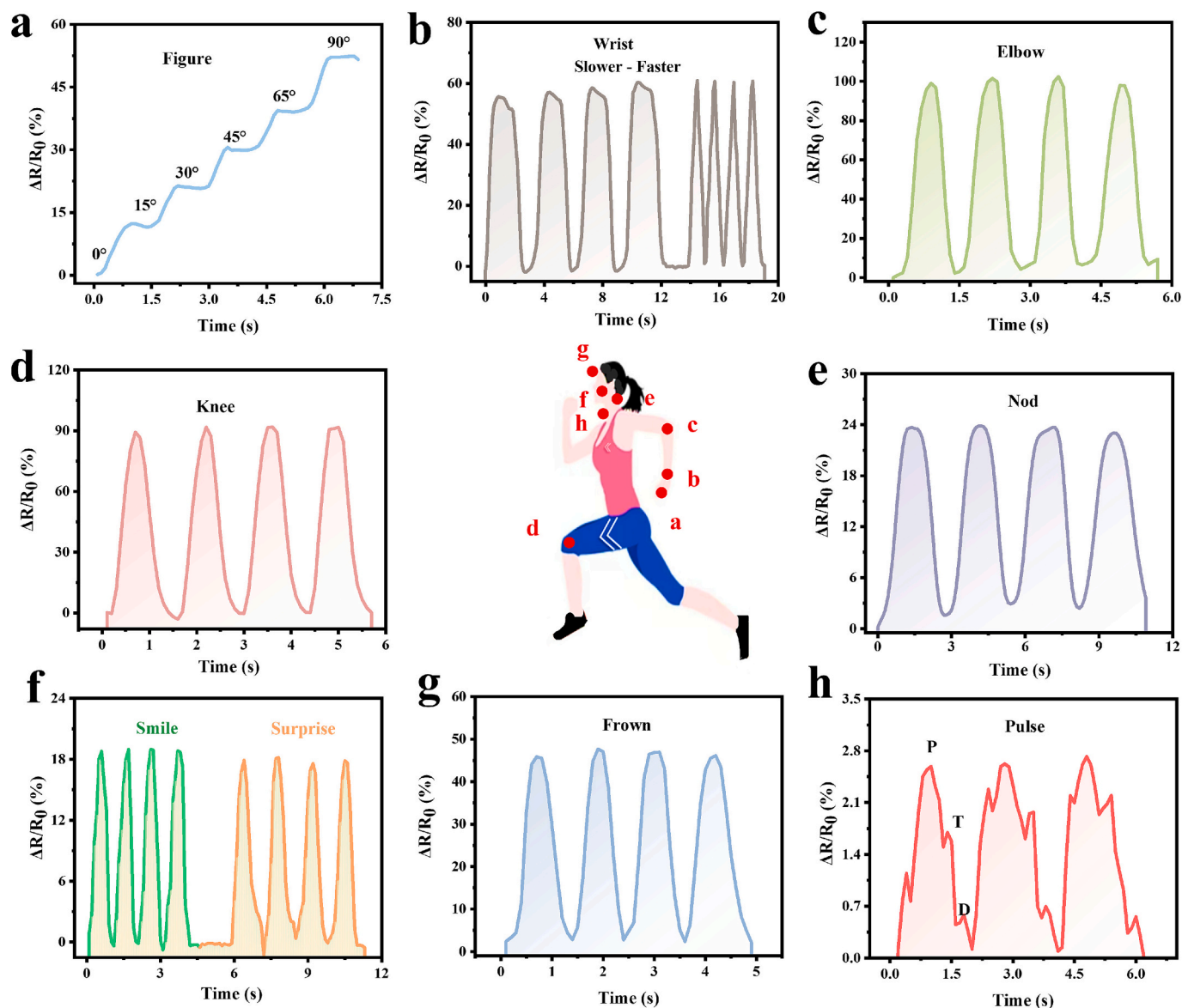


Fig. 6. Relative resistance changes of the hydrogel sensor during human activity monitoring. (a) Different finger bending angles (0–90°). (b) Different wrist bending rates. (c) Elbow. (d) Knee. (e) Nodding. (f) Facial expressions. (g) Frowning. (h) Pulse.

knee, and neck (Figs. 6b–e), but also possessed the high sensitivity required in the field of human motion detection. In addition, when the sensor was attached to the facial and forehead skin respectively, it could effectively recognize rapid emotional changes by capturing subtle deformation signals associated with facial microexpressions and emotional fluctuations (e.g., surprise, smiling) (Figs. 6f–g). More importantly, the amplified wrist pulse signals showed characteristic peaks related to blood pressure and heart rate, specifically including the percussion wave (P), tidal wave (T), and diastolic trough (D) (Fig. 6h). In conclusion, the hydrogel strain sensor could facilitate the monitoring and prediction of an individual's physiological status, and had potential applications in clinical diagnosis and disease prevention.

Applications of the hydrogel pressure sensor. The hydrogel pressure sensor exhibited high compressibility, excellent resilience, and high pressure sensitivity. As shown in the schematic diagram of Fig. 7a, different writing contents exerted different stresses on the sensor sliding path, thereby inducing the sensor to generate distinct response signals. As a qualitative proof-of-concept, both numbers (“1”, “2”, “3”) and letters (“A”, “B”, “C”) written on a flexible board could trigger the sensor to output unique and reproducible signal patterns (Fig. 7b and c),

qualitatively demonstrating its ability to distinguish different writing inputs. Furthermore, the excellent pressure sensitivity enabled basic information transmission via Morse code, where a light touch generated a “dot” signal and a press generated a “dash” signal (Fig. 7d). Fig. 7e–g illustrate the transmission of the messages “SOS”, “NO”, and “ABC”, respectively, where different combinations of dots and dashes were recorded in real time by adjusting the pressing force. These preliminary results highlight the potential of the hydrogel sensor for human-machine interaction, although it should be noted that the current demonstrations are qualitative. Comprehensive quantitative assessment of recognition accuracy, signal-to-noise ratio, and anti-interference capability in a complete system remains an important direction for future work.

4. Conclusion

This study proposes a Bet-NaCl synergistic enhancement strategy, successfully preparing a multifunctional CSMA-AM@Bet-NaCl hydrogel. Based on this hydrogel, a high-performance flexible strain/pressure sensor was constructed, providing a novel technical pathway for the design of flexible sensing materials. The introduction of this synergistic

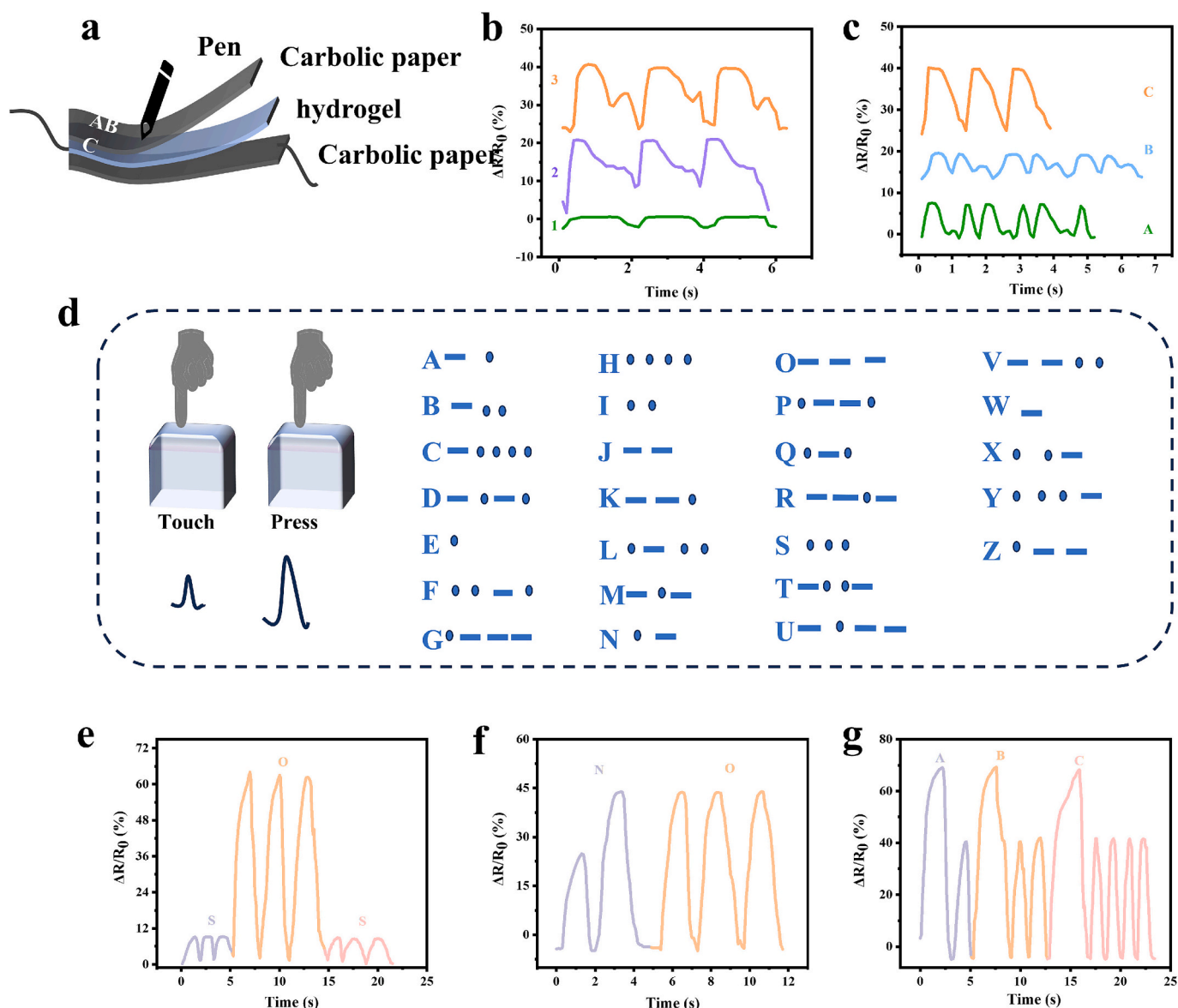


Fig. 7. Applications of the hydrogel pressure sensor. (a) Schematic diagram of writing with the hydrogel. (b-c) Word recognition for handwriting monitoring (1, 2, 3, A, B, C). (d) Morse code decoding table in alphabetical order and signal conversion of the hydrogel sensor for Morse codes "Touch" and "Press". (e-g) $\Delta R/R_0$ signal communication via Morse code, including (e) "SOS", (f) "NO", and (g) "ABC".

strategy achieves synchronous improvement in the electrical and mechanical properties of the hydrogel: the prepared hydrogel exhibits a high conductivity of 5.56 S/m, excellent toughness of 6.99 MJ/m³, and high compressive strength of 6.68 MPa under 90% compressive strain, breaking through the bottleneck that traditional hydrogel sensing materials have difficulty balancing electrical and mechanical properties. The sensor based on this hydrogel demonstrates excellent sensing performance, with a sensitivity of $GF = 3.18$ in the 250% ~ 500% strain range and $S_1 = 0.642 \text{ kPa}^{-1}$ in the 0–50 kPa pressure range; simultaneously, the hydrogel has efficient self-healing ability, with a self-healing rate of up to 96.6% within 16 h, enabling rapid functional recovery after damage. The sensor can stably and accurately monitor large deformation movements of human joints, as well as subtle strains caused

by facial micro-expressions and vocalization, showing great application potential in wearable health monitoring, human-computer interaction, and other fields.

CRediT authorship contribution statement

Yanju Zhang: Software, Investigation, Formal analysis, Data curation, Conceptualization. **Yuxin Mao:** Investigation, Data curation. **Chenguang Yan:** Formal analysis, Data curation. **Xudong Yu:** Writing – review & editing, Visualization, Validation, Supervision, Funding acquisition, Conceptualization. **Yajuan Li:** Writing – review & editing, Visualization, Validation, Supervision, Investigation, Data curation, Conceptualization.

Declaration of competing interest

There are no conflicts of interest to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijbiomac.2026.153221>.

Data availability

Data will be made available on request.

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